

Received 26 September 2025, accepted 13 October 2025, date of publication 22 October 2025, date of current version 30 October 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3624284

 SURVEY

A Comprehensive Review of Deepfake Detection Techniques Utilizing Remote Photoplethysmography

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This work was supported by Amrita Vishwa Vidyapeetham.

ABSTRACT Deepfake technology enables the creation of highly realistic but fabricated videos, raising significant concerns about the authenticity and reliability of digital media. As deepfakes become increasingly sophisticated, effective detection methods have become crucial. One promising approach involves Remote PhotoPlethysmography (rPPG), a non-contact method that measures subtle changes in skin color to estimate heart rate. rPPG is a powerful tool for detection as deepfake videos often fail to replicate the physiological signals inherent in real human subjects. This survey aims to provide a comprehensive understanding on various rPPG-based techniques for identifying deepfakes and analyzing their effectiveness, challenges, and future potential. To improve the detection accuracy, methods that enhance rPPG signals in combination with machine learning models are examined. Real-world applications, including the detection of fake celebrity videos that are employed in propaganda, illustrate the applicability of these techniques. Through a comprehensive review of existing rPPG-based deepfake detection techniques, this survey aims to inform future research and development endeavors in the area, thereby contributing to broader efforts to secure digital content from tampering and exploitation.

INDEX TERMS Deepfakes, deepfakes generation, deep learning, GANs, rPPG, VAEs.

I. INTRODUCTION

Deepfake has increasingly become a digital threat in the recent years. With the help of tools like AI and ML, a person's likeness towards something is manipulated, therefore generating hyper-realistic media called Deepfake. An untrained eye, cannot detect these deepfake-edited movies, photos, and audio recordings that can look realistic. Deepfake technology poses ethical, legal, and security problems by swapping faces, modifying speech, and producing false information.

The associate editor coordinating the review of this manuscript and approving it for publication was Yeon-Ho Chung^{id}.

Deepfakes may propagate falsehoods, harm reputations, and aid identity theft by effectively mimicking actual people. In other words, synthetic media in which a person's image is digitally changed or replaced using artificial intelligence are referred to as deepfakes, a combination of deep learning [1] and fake.

Deepfakes' widespread consequences need immediate action. The integrity of the information is at risk as deepfakes become more sophisticated and common. The political usage of deepfakes can influence public opinion, undermine democracy, and cause social unrest. AI has recently generated and screened deepfake videos of political

leaders. These technologies affect public trust and political stability by using deepfakes to disseminate misleading information and distorting voter attitudes [2]. Deepfakes may emotionally and psychologically traumatize victims of harassment, cyberbullying [3], and pornographic reprisal [4]. Deepfakes undermine justice by also eroding digital evidence in court. These concerns make efficient detection and mitigation techniques essential to protect society as well as individuals from deepfake technology.

Popular deepfake and face-swapping tools include Reface [5], MyHeritage [6], DeepFaceLab [7], and DeepFaker App [8]. Realistic face swaps in movies and GIFs may be implemented quickly using the entertaining Reface software and the user-friendly Deepfakes internet platform. Through the “Deep Nostalgia” function on MyHeritage, users can animate vintage images, bringing historical figures to life. While the DeepFaker app’s intuitive design allows novices to create deepfakes, DeepFaceLab’s sophisticated capabilities attract academics and enthusiasts.

Advanced deep learning techniques such as generative adversarial networks (GANs) [9] and variational auto-encoders (VAEs) [10] are used to make Deepfakes. These more complex models, like GANs with their two generation and discriminator parts, are very good at using false representations that are very close to the real representation to make obviously false stuff, all while constantly checking their work against real data. Autoencoders have also made it possible to rebuild and shrink pictures, which lets us create new faces that look real from smaller ones. This is easier to utilize when the quantity of the files with images and videos of the target group is massive. Deepfakes require intensive training to accurately replicate facial expressions and features. Recent research [11] indicates the growing maturity of deepfake generation with advanced deep learning architectures like GANs and VAEs, emphasizing the necessity of AI-driven detection techniques to counter their deceptive credibility.

After the training process with a broad set of scenarios, algorithms like the Advanced face-swapping algorithm make it easy for such models to be able to easily spot the features of the targeted individual [12]. This allows users to tap into their own creativity to create creative media. Multiple processes of post-processing have been employed to change colors and smoothen out any imperfections in the content; also, we provide details to make it look realistic. Deepfakes proved to be extremely effective, but since they are used so often that, it would be difficult to distinguish them and stay away from them. As the process of deciding whether a person is lying becomes increasingly difficult it is essential to have functional solutions.

Due to the evolution of the deepfake-generation methods and the convincing nature of these fakes, it is becoming increasingly challenging to detect deepfakes. Many solutions to this challenge have been proposed by researchers and engineers using advances in computer vision, machine learning, and digital forensics [13], [14], [15]. Seeking physiological

and facial clues is a common way to identify discrepancies indicating deepfake manipulation [16]. An alternative promising strategy is to develop deep learning-based detection algorithms trained on large datasets of real and synthetic media [17]. Researchers have investigated algorithmic and forensic analytic techniques to detect proof of manipulation in digital media [18]. One of these strategies is to examine metadata, digital artifacts from the deepfake manufacturing process, and picture compression artifacts. Also, changes to blockchain technology has made it possible for digital signatures to be tampered with. These signs can be used to verify the accuracy of video data and to demonstrate that any unauthorized changes have been made [19].

Deepfake manufacturing technologies, on the other hand, are making it much harder to study how to find deepfakes [20]. Therefore, people from different fields will have to work together and keep developing new ideas to propose reliable, scalable algorithms that can find deepfake content and stop it from spreading.

This survey aims to comprehensively examine and assess various deepfake detection techniques that leverage rPPG. The survey aims to evaluate the effectiveness of these methods in distinguishing between real and manipulated content, considering their accuracy, robustness, and applicability across different scenarios. With deepfakes being developed with the help of advanced machine learning and deep learning methods that generate very accurate synthetic media, rPPG-based detection has evolved as a sophisticated approach, moving beyond early detection methods focused on visual artifacts. This survey will also identify key challenges and limitations associated with rPPG-based detection techniques and suggest potential avenues for future research to improve their performance in the evolving landscape of deepfake technology.

rPPG is a recent non-contact technique that enables the extraction of physiological signals such as heart rate, which can be detected from peripheral video cameras using facial images. This method, which has received considerable attention in the biomedical and computer vision communities, offers a new way to monitor vital signs without physical touch [21], [22]. Lately, the application of rPPG has expanded to the medical field to quantify blood oxygen saturation and heart rate [23] and to prevent spoofing [24].

In the current landscape of deepfake detection, most traditional methods rely on pixel-based features, visual inconsistencies, or audio-visual alignment. However, these approaches are increasingly vulnerable to the advances in generative models that create nearly flawless visual and temporal consistencies. In contrast, rPPG leverages physiological signals—specifically the subtle variations in skin color due to the flow of blood—that are inherently difficult to replicate with high fidelity in synthetic media. This is because deepfake algorithms typically focus on facial features and texture synthesis while ignoring the complex, temporally coherent biological rhythms embedded in real

human faces. As a result, fake videos often exhibit unnatural or absent pulse signals, inconsistent heart rate variability, and uniform rPPG distribution across facial regions. These artifacts are largely invisible to human perception but can be extracted and analyzed through rPPG signal processing. Therefore, rPPG offers a fundamentally different and more resilient signal domain for detecting deepfakes. It acts as a physiological watermark—something real humans possess certain characteristics but synthetic avatars often fail to mimic them accurately. By focusing on rPPG, this survey addresses an emerging detection modality that could serve as a cornerstone of future multimodal detection frameworks.

In the current generation, deepfakes have been considered a significant threat to content, as they are created using artificial intelligence. Although the rPPG of the body can sense slight changes in the physiological response of a person, the same signals introduce a new angle to improve the deepfake detection approach, leveraging physiological regularities to enhance generalization of diverse datasets and improve detection robustness beyond pixel-domain features [25]. As can be seen, rPPG can detect pulsatile blood flow and differentiate between artifacts that emerge during deepfake generation and are invisible to the human eye [26].

Recent studies have confirmed that rPPG-based techniques are reliable in spotting authentic or fake videos, primarily using blood flow rhythm and physiological signals. This places rPPG in an advantageous position to assist in developing new deep learning based techniques for the next-generation deepfake identification of improved accuracy and reliability.

The survey paper is organized as follows: Section II presents a Literature Review, summarizing relevant research in deepfake detection. Section III discusses the Advancements of Computational Methods in Deepfake Generation, focusing on recent breakthroughs in synthetic media creation. Section IV covers the Advancements in deepfake detection techniques, outlining innovations in detecting fake content. Section V introduces rPPG, a technique for extracting physiological signals from video. Section VI explores the Types of rPPG Methods in Deepfake Detection. Section VII addresses the challenges in deepfake detection using rPPG. Section VIII looks at future directions for enhancing deepfake detection. Finally, Section IX concludes the paper.

II. OVERVIEW OF LITERATURE REVIEW

Deepfake detection has gained significant attention in recent years due to the rapid advancements in AI and Deep Learning (DL). The increasing realism of deepfake media poses ethical, legal, and security threats, necessitating robust detection mechanisms. Several approaches have been explored in the literature, including traditional image forensics, deep learning-based detection, physiological signal analysis, and hybrid techniques. This section classifies and reviews the various deepfake detection methods, focusing on their effectiveness and contributions to the field.

Traditional forensic methods focus on detecting inconsistencies in images and videos using handcrafted features. These approaches include pixel-level analysis, which identifies anomalies such as unnatural blurring, pixelation, and inconsistencies in shadows and reflections [27]. Compression artifact detection examines differences in compression patterns between the original and manipulated content to identify tampered areas. Metadata and error-level analysis focus on detecting inconsistencies and artifacts left during the deepfake creation process. Despite their effectiveness in certain cases, these methods often fail when faced with advanced deepfake generating methods that are capable of minimizing such artifacts. Models such as MesoNet4 and ResNet-101 [28] and Visual Geometry Group (VGG-19) [29] have been utilized for pixel-level analysis and deepfake forensic detection. However, the VGG16 model can only extract the image features, but for these techniques to be successful, classification is also required. An additional model such as ResNet or CNN should be incorporated for this purpose.

Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have immensely contributed to deepfake detection. These methods concentrate on detecting intricate patterns that assist in distinguishing between real and fake media. To scan deepfake videos for spatial inconsistencies CNNs are employed. Networks like XceptionNet have been trained on datasets like FaceForensics++ (FF++) for classifying deepfakes with high accuracy [30], [31]. RNNs and Long Short-Term Memory (LSTM) Networks study temporal inconsistencies in facial motion and blinking behavior since deepfake videos tend to lack natural motion coherence. New models like Vision Transformers (ViTs) extract long-range dependencies in facial features, improving the robustness of detection. Notable models include Interpretable Spatial-Temporal Video Transformer (ISTVT) [32], Convolutional Vision Transformers (CViT) [33], and Hybrid CNN-LSTM models [34], which have demonstrated strong performance in detecting deepfake content.

To improve detection accuracy, multi-modal techniques integrate different modalities such as audio, textual, and visual data. Audio-visual synchronisation analysis checks if lip movements match speech audio, as deepfake videos often exhibit misalignment [35]. Recently, a prominent watermarking technique based on GANs was developed to detect proactive facial manipulations [36]. Fusion-based approaches combine multiple detection methods, such as fusing rPPG-based physiological signals with CNN-based visual analysis, to enhance robustness against sophisticated deepfakes. Advanced multi-modal detection models such as EfficientNetB4 and Time-Delay Neural Network (TDNN) [31], Vision-Language Pretraining with Gate Fusion (VLP-GF) [35], and FakeCatcher [37] have been designed to integrate various modalities for improved deepfake detection.

rPPG is a non-contact method used to measure subtle changes in skin color due to blood flow. Heart rate and pulse signal analysis extracts heartbeat signals from facial videos

by detecting periodic variations in skin tone [38]. Several studies have shown that deepfake videos lack consistent heart rate signals, making them distinguishable from real videos. Deepfake videos frequently do not accurately reproduce these physiological signals, rendering rPPG-based identification a viable method. The features derived from the rPPG-based on the signal provide explicit information regarding the physiological behavior of heart rate, rendering it an efficient tool for deepfake identification [39]. Hybrid rPPG models combine rPPG with CNNs or ViTs to enhance deepfake detection by integrating both physiological and visual inconsistencies. Thus, models such as DeepFakesON-Phys [38], simple rPPG-based methods [39], and DeepRhythm [40] have demonstrated effectiveness in utilizing physiological signals for deepfake detection.

Several datasets have been developed to train and evaluate deepfake detection models. FF++ [30] contains manipulated videos using various deepfake generation methods, serving as a benchmark for CNN-based detection. Celeb-DF [41] is a dataset of high-quality deepfake videos designed to challenge current detection models. DFDC (DeepFake Detection Challenge Dataset) [42], developed by Facebook AI in collaboration with academic and industry partners, includes diverse deepfake videos with varying degrees of realism. VIPL-HR [43] is designed explicitly for evaluating rPPG-based deepfake detection techniques.

This literature review classifies and analyses deepfake detection techniques, emphasizing rPPG-based approaches. The integration of physiological signal analysis with deep learning methods offers a promising direction for improving detection accuracy and robustness. By leveraging multi-modal techniques and continuously updating datasets, researchers can enhance the ability to combat deepfake threats effectively.

III. ADVANCEMENTS OF COMPUTATIONAL METHODS IN DEEFAKE GENERATION

A. INTRODUCTION TO DEEFAKE GENERATION

Deepfakes are synthetic media that involve substituting the source person's image with another, achieved by means of complex neural networks like GANs and VAE. Exploration of deepfakes and their potential threat to face recognition, along with initial detection strategies, was performed in [44]. Further works provided explanations of how different architectures function [45].

Understanding the computational methods used to create deepfakes is crucial, as it enables the establishment of reliable mechanisms for building confidence in digital media and helps formulate policies and laws to guide against misuse [46]. Table 1 presents a comparative study of various computational methods for deepfake generation [1], [7], [8]. Moreover, the current deepfake content often lacks coherent rPPG-based physiological signals, providing a robust cue for detection and linking this study to physiological deepfake forensics.

B. EARLY METHODS IN DEEFAKE GENERATION

Historical Context: Deepfake technology originated from the development of computer graphics and digital image processing. Early face swaps were performed manually using basic image processing, producing low-quality and unrealistic outputs.

The breakthrough came with machine learning and neural network algorithms in the 2010s, where Eigenfaces and Principal Component Analysis (PCA) were applied for face recognition and swapping, though these methods relied on handcrafted features and had low generalisation across datasets [47].

C. BASIC ALGORITHMS AND CHALLENGES

Early face-swapping methods used geometric transforms and color adjustments to map source faces onto target images. However, these methods did not capture any coherent rPPG-based physiological signals, making them inherently detectable through physiological cues.

1) GEOMETRIC WARPING AND COLOR CORRECTION

Corresponding feature points on the source and target images were detected to map the faces, and geometric transformations were applied to deform the source face to the target shape. Technical preprocessing adjusted skin tones and lighting conditions between source and target faces. Such techniques often left tell-tale signs, including seams, inconsistent skin tones, and unrealistic textures, with low resolution [48].

2) LIGHTING AND EXPRESSION HANDLING

These methods were ineffective in dealing with illumination and pose variations, causing temporal discrepancies and flicker effects in videos [49].

D. ADVANCEMENTS IN GAN ARCHITECTURES

Generative Adversarial Networks (GANs) are foundational techniques in deepfake generation, consisting of a generator that creates synthetic data and a discriminator that distinguishes between real and fake data. This adversarial training aims to produce increasingly realistic outputs, particularly effective in face swapping and high-quality image synthesis [50]. Training on large datasets of target and source faces allows GANs to learn expressions, poses, and lighting variations to generate deepfake videos.

The core mechanism of GANs lies in simultaneously refining the generator to deceive the discriminator, enabling the model to learn complex data distributions. The generator and discriminator loss functions are defined as follows [51]:

$$\mathcal{L}_G = \log(1 - D(G(z))) \quad (1)$$

$$\mathcal{L}_D = -\log(D(x)) - \log(1 - D(G(z))) \quad (2)$$

The combined min-max objective is [50]:

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (3)$$

Early improvements like DCGAN employed convolutional layers with ReLU in the generator and LeakyReLU in the discriminator to improve image realism. Despite enabling high-resolution image generation, DCGANs faced limitations such as mode collapse and training instability [52].

Progressive Growing of GANs (ProGAN) stabilized training by starting with low-resolution images and incrementally increasing to high resolution. This hierarchical learning allowed the model to capture coarse-to-fine features and achieve state-of-the-art quality [53], [54].

StyleGAN and StyleGAN2 further advanced realism using style-based generators and AdaIN for attribute control, improving image sharpness and reducing artifacts. These architectures remain state-of-the-art for deepfake generation [55], [56].

While GANs can now produce photorealistic faces with accurate expressions and lighting, they do not reproduce coherent physiological signals such as rPPG-based blood flow patterns. This absence of physiological consistency provides a critical cue for robust deepfake detection using rPPG.

E. OTHER COMPUTATIONAL TECHNIQUES

1) AUTOENCODERS

Autoencoders are neural networks that learn efficient input data coding, consisting of an encoder that maps the input to a latent representation and a decoder that reconstructs the input from that representation. In deepfake generation, they are primarily used for face swapping, where an input face is encoded into a latent vector and reconstructed as another individual's face.

Their limitation lies in the fixed-dimensional input and output, making it challenging to handle varying facial orientations and expressions. Despite this, autoencoders laid the foundation for deepfake generation by providing a framework for encoding and decoding facial data efficiently. However, autoencoder-generated deepfakes lack natural physiological coherence, particularly rPPG-based signals, which can be leveraged for detection.

2) VARIATIONAL AUTOENCODERS (VAES)

VAEs extend autoencoders by introducing probabilistic latent representations, allowing controlled and diverse image generation. In deepfake applications, VAEs interpolate in the latent space to synthesize non-existent realistic faces by combining features from multiple sources [57].

Unlike standard autoencoders that produce a single deterministic output, VAEs sample from a Gaussian latent distribution, enabling the generation of multiple plausible variants of the same individual. This makes VAEs suitable for creating diverse instances of a person's face under varying conditions.

The ELBO objective function, which VAEs optimize, is given in Equation 4 [10]:

$$\mathcal{L}_{\theta, \phi}(\mathbf{x}) = \log p_{\theta}(\mathbf{x}) - D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p_{\theta}(\mathbf{z}|\mathbf{x})) \quad (4)$$

where: $q_{\phi}(\mathbf{z}|\mathbf{x})$ is the approximate posterior distribution (encoder), and D_{KL} is the Kullback-Leibler divergence from the prior $p(\mathbf{z})$.

While VAEs generate diverse and realistic outputs, they do not replicate the temporally coherent blood-flow patterns visible in rPPG signals, making physiological analysis an effective detection cue.

3) RECURRENT NEURAL NETWORKS (RNNs)

RNNs are designed to process sequential data and are particularly suitable for audio and video deepfakes that require temporal consistency. By retaining information from previous steps, RNNs can produce sequentially coherent frames or audio segments for deepfake synthesis [58].

Their primary limitation is the difficulty in maintaining long-term dependencies, leading to degradation in performance for lengthy sequences. Nevertheless, RNNs have been successfully used in generating realistic speech and video sequences.

Even though RNN-based deepfakes capture short-term temporal dynamics, they still fail to produce the subtle, periodic variations in skin tone caused by blood circulation. These rPPG-based cues can therefore reveal the synthetic nature of such deepfakes, providing a robust detection approach.

F. DEEP LEARNING FRAMEWORKS AND TOOLS

Deep learning frameworks are essential for implementing and training deepfake generation models, with TensorFlow, PyTorch, and Keras being widely used. TensorFlow [59], an open-source framework, provides high-level APIs for model creation and deployment, enabling researchers to explore architectures and optimize workflows. PyTorch [60] is valued for its dynamic computation graph, usability, and libraries like Hydra that support iterative debugging. Keras [61], built on TensorFlow, simplifies deep learning with an intuitive design and fast model construction.

Specialised tools such as SimSwap++ [62], FaceSwap [63], and DeepArt complement these frameworks. SimSwap++ integrates adaptive identity preservation with a multi-attribute encoder for high-quality face swaps. FaceSwap supports detection, alignment, and realistic merging, while remaining beginner-friendly. DeepArt, primarily for artistic style transfer, demonstrates deep learning's creative potential and improves the quality of synthetic outputs. Together, these frameworks and tools streamline deepfake generation and broaden accessibility.

G. DEEFAKE GENERATION USING RPPG SIGNALS

rPPG is effective in synthesizing physiological signals like pulsating signals from facial videos, thus making it possible to synthesize deepfakes. Due to changes in blood flow that affect skin tone, rPPG allows the capture of photoplethysmographic signals from video for deepfake generation pipelines. This integration enables one to generate

synthetic videos that mimic the target subject in both looks and even the physiological one, making it even harder to detect them [64].

Improving deep learning and computer vision enables the generation of deepfakes with rPPG signal incorporation. Advanced neural networks can analyze facial videos and produce corresponding rPPG signals aligned with the video content to create physiologically realistic deepfakes [65]. Babaei et al. highlighted how such advancements complicate traditional detection methods in the context of generative AI [66]. This approach is particularly relevant in medical simulations and digital modeling of human behavior [67].

However, generating realistic rPPG signals faces challenges under varying lighting, facial expressions, and video quality. Future research focuses on enhancing the reliability of rPPG-based deepfake generation to ensure physiologically consistent outputs.

IV. EVOLUTION OF DEEPFAKE DETECTION TECHNIQUES

A. EARLY DETECTION METHODS

1) EYE MOVEMENT

Early deepfake detection relied on gaze direction and blink rates, as these natural eye behaviors are challenging to replicate in synthetic media [68]. Although effective in some cases, these methods are vulnerable to improvements in generative models and do not consider physiological signals such as rPPG, which provide a more resilient cue for detection.

2) VISUAL ARTIFACTS

Another early approach identified inconsistencies in video textures, including flickering, unnatural skin tones, or uneven lighting and shadows. Such artifacts served as key indicators for marking probable fake content [69]. However, as visual quality improves, these cues become less reliable compared to physiological analysis based on rPPG signals.

3) PIXEL AVERAGING OF THE HUMAN FACE

Pixel-level analysis compares the distribution of pixels in suspicious faces to genuine ones, identifying disparities that indicate manipulation [70]. While useful for early detection, pixel analysis does not capture subtle blood-flow variations that rPPG can reveal, offering a deeper verification layer.

B. MACHINE LEARNING AND DEEP LEARNING APPROACHES

ML and DL techniques analyze large datasets to identify patterns distinguishing real from manipulated content. With more data, these models improve in accuracy and robustness. Detection is commonly performed using Convolutional Neural Networks (CNNs) and temporal analysis, with methods summarized in Table 1 and further detailed in Table 2.

1) CONVOLUTIONAL NEURAL NETWORKS (CNNs)

CNNs learn spatial features from images to identify defects in deepfakes, such as uneven textures, unrealistic lighting, or blending artifacts [27]. They process frames through convolutional filters, pooling, and fully connected layers to extract hierarchical features and classify content as real or fake. Trained on large datasets of authentic and synthetic images, CNNs can detect manipulations such as face swaps, expression changes, and video synthesis. Pixel-level variations around sensitive regions like lips and eyes—where deepfake algorithms struggle—are key to their effectiveness [71].

Despite their success, CNNs focus on visual cues and remain vulnerable to next-generation deepfakes that exhibit high spatial fidelity. Integrating rPPG signals into CNN-based pipelines enhances robustness, as rPPG captures subtle, periodic skin-color variations linked to real blood circulation—patterns nearly impossible to replicate in synthetic media.

2) TEMPORAL ANALYSIS

For video-based deepfakes, temporal analysis examines how features persist across frames, identifying inconsistencies that arise because frame-to-frame synthesis is challenging. Deepfakes often exhibit sudden lighting changes, unnatural motions, and inconsistent facial expressions [72]. Tracking facial landmarks or analyzing eye and lip movements across frames can reveal subtle manipulations.

The process involves breaking a video into frames and examining transitions between them. Motion estimation techniques evaluate pixel flows to verify whether they follow natural patterns [73]. Combining spatial and temporal analysis strengthens deepfake detection, as it captures anomalies that static frame analysis may miss.

C. ADVANCED INTEGRATIONS

1) MULTI-MODAL DETECTION

As deepfake technology has advanced, more comprehensive detection methods have become essential. Multi-modal detection integrates audio, visual, and textual cues to improve accuracy [74]. For instance, subtle mismatches between lip movements and audio or inconsistencies in speech patterns can expose deepfakes. By cross-referencing multiple data streams, multi-modal systems capture anomalies that single-modal methods may overlook, particularly in high-quality fakes designed to bypass visual-only detectors [75]. Integrating rPPG into multi-modal detection enhances robustness by adding a physiological verification layer. Even if visual and audio cues appear authentic, a lack of coherent blood-flow signals in the face expose deepfakes.

2) GANS FOR DETECTION

GANs, used initially to generate deepfakes, have also been repurposed for detection. In this context, the discriminator is trained to identify subtle differences between real and

TABLE 1. Comparison of computational methods for deepfake generation.

Method	Advantages	Disadvantages
Autoencoders [10]	- Easy to implement and understand, making them accessible for beginners. - Effectively reduces data dimensionality, useful for feature learning and data preprocessing.	- The deterministic nature of standard autoencoders constrains their ability to produce diverse and highly realistic outputs.
Variational Autoencoders [10]	- By learning probability distributions, VAEs can capture more nuanced representations of data. - The probabilistic approach allows for sampling from learned distributions, producing varied results.	- VAEs often produce blurrier or less detailed images due to the probabilistic encoding and decoding process.
Recurrent Neural Networks [10]	- Excels in processing and generating time-series data, making them suitable for video and audio deepfakes. - Can maintain context over time, crucial for coherent video generation.	- RNNs are not specifically designed for image tasks, leading to lower-quality visual outputs compared to specialized image generation models.
GANs [10]	- The adversarial training process pushes the generator to produce increasingly convincing outputs.	- The balance between generator and discriminator can be difficult to maintain. - The generator may focus on a limited set of outputs that fool the discriminator, reducing diversity.
Deepfacelab [7]	- Offers optimized implementations of neural network architectures and training algorithms.	- Effective training often necessitates substantial data and computational power, which can be costly and time-consuming.
Deepfaker [8]	- Incorporates heart rate and other vital signs into deepfakes, enhancing realism. - Can work with lower resolution videos as it focuses on physiological cues.	- Movements can interfere with accurate signal detection. - Requires high-quality video and advanced processing: needs specialized equipment and algorithms for accurate physiological signal extraction.

synthetic content [76]. Through adversarial training, the discriminator continuously improves as generators attempt to produce more convincing fakes [77]. This dynamic push-and-pull ensures detection models remain effective against evolving deepfakes.

Notable applications include Reality Defender, which analyzes video calls in real-time to detect deepfakes [78]. GAN-based detection models can also be fine-tuned for specific manipulation types, including face swaps, voice alterations, and full-body avatars, making them highly adaptable for content moderation and forensics. Despite this adaptability, most GAN-based detectors rely on visual or statistical cues alone. Incorporating rPPG signals into GAN detection frameworks can enhance reliability, as physiological consistency is extremely difficult to falsify in generative models.

D. SUITABLE DATASETS FOR DEEFAKE DETECTION

1) FACEFORENSICS++ DATASET

A widely used dataset for evaluating deepfake detection algorithms is FaceForensics++ (FF++) [30]. This dataset includes videos that are both authentic and altered, which are produced by techniques like DeepFakes, Face2Face, FaceSwap, and Neural Textures. This dataset acts as a reliable standard for creating and evaluating models, as these videos come from authentic contexts like interviews. The absence of physiological signals like rPPG in FF++ hinders its applicability for creating the next-generation physiological detection techniques, despite the fact that it offers a wide range of visual manipulations.

2) CELEB-DF DATASET

The Celeb-DF dataset targets high-quality deepfakes that are challenging to detect [41]. It consists of 5,639 videos

featuring 59 celebrities, with 590 real and 5,049 fake videos. Its emphasis on realistic manipulations makes it a critical benchmark for developing models capable of handling professional-grade content. However, like FF++, Celeb-DF focuses purely on visual realism without providing physiological ground-truth data. This gap highlights the need for datasets that include rPPG signals to support physiologically aware deepfake detection.

Overall, advanced detection methods and datasets have significantly improved deepfake detection but remain primarily reliant on visual and auditory cues. This limitation motivates the exploration of Remote Photoplethysmography (rPPG), which leverages subtle skin color variations from blood flow as a physiological watermark for detecting synthetic media. The following section introduces rPPG, its principles, and its potential in deepfake detection.

V. INTRODUCTION TO REMOTE PHOTOPLETHYSMOGRAPHY

A. EARLIER METHODS FOR MEASURING PHYSIOLOGICAL FEATURES

Comprehending physiological characteristics is essential for the diagnosis, monitoring, and treatment of various health disorders. Conventional techniques, such as pulse oximetry [79], Doppler ultrasonography [80], and electrocardiography (ECG) [81], have historically served as the foundation of clinical practice, providing vital insights into respiratory and cardiovascular health. The ECG, introduced in the early twentieth century, revolutionised diagnostics by giving a non-invasive and cost-effective method of detecting arrhythmias and ischemic heart diseases. ECG is now a critical tool in clinical settings, and it is used in stress testing and intensive care units. For continuous remote monitoring of heart issues it has been integrated into telehealth systems.

TABLE 2. Overview of deepfake detection techniques, datasets, and their analysis.

Model	Technique	Datasets	Advantages	Disadvantages	Results and Analysis
MesoNet4 and ResNet-101 [28]	Eye Movement: Hybrid deep learning approach combining MesoNet4 and ResNet-101 for facial feature analysis	FaceForensics++, Celeb-DF v1, Celeb-DF v2	High accuracy in detecting subtle visual cues (e.g., lighting inconsistencies, facial motion) also robust and adaptable to lighting/environment variations.	Limited real-time detection in live-streaming contexts	Achieved an accuracy of 98.73% on FaceForensics++, 96.89% on Celeb-DF V1, and 97.90% on Celeb-DF V2 for Deepfake detection.
ISTVT [32]	Decomposed spatial-temporal self-attention with self-subtract mechanism, transformer-based architecture	FaceForensics++, FaceShifter, DeeperForensics, Celeb-DF, DFDC	Enhanced detection via spatial-temporal decomposition, interpretable visualisation of discriminative regions, strong generalisation across datasets.	Higher computational complexity due to transformer architecture, performance may vary with dataset diversity.	Achieved estimated 99.8% accuracy on Celeb-DF, 92.1% on DFDC
VGG-19 Deepfake Detection Model [29]	Pixel-Level Analysis with Deep Learning	140K Real and Fake Faces Dataset (contains GAN-generated and expert-generated images)	High accuracy in distinguishing real and fake images; robust feature extraction using deep CNN layers; effective in identifying GAN-generated traces.	Computationally expensive; requires large-scale training data; potential for overfitting.	Achieves 95% accuracy with F1-score of 95%. Precision of 93%, Recall of 97%, and ROC-AUC of 96%. PCA-SVM classification improved clustering of real and fake images.
CViT [33]	The model applies CNN for feature extraction (eyes and nose) and CViT for face detection, with majority voting for final predictions.	FaceForensics++, DFDC datasets.	High accuracy, robustness against deepfake manipulations, and improved feature extraction	Dependency on specific datasets, and potential inefficacy in detecting unseen deepfake variations	Achieved 96.98% accuracy for the eye region, 96.76% accuracy for the nose region, and 85% accuracy for the face region.
Hybrid CNN-LSTM [34]	A frequency-based frame extraction method is applied, followed by CNN and LSTM-CNN for deepfake video detection.	FaceForensics++, DFDC.	Improved feature learning and effective for sequence prediction in videos.	Lower accuracy on DFDC compared to FF++ and performance depends on training-testing split	LSTM-CNN Achieved highest 82% accuracy on FF++ (80-20 split) and 72% on DFDC (both 80-20 and 70-30 split).
EfficientNetB4 and Time-Delay Neural Network (TDNN) [31]	Multi-Modal: Combines audio and visual features to detect inconsistencies across modalities.	FaceForensics++, FakeAVCeleb, DFDC, VidTIMIT, DeepfakeTIMIT, Faceswap, ASVspoof 2019, TIMIT-TTS.	Demonstrates high generalization capabilities on unseen multimodal data and trained on disjoint monomodal datasets.	Requires high-quality monomodal datasets and fusion of modalities and temporal analysis increases computational demands.	Achieved high accuracy (up to 94.6% on DFDC), with mid-fusion outperforming other strategies by effectively combining audio-visual features for robust detection.
Vision-Language Pretraining with Gate Fusion (VLP-GF) [35].	Multi-Modal: combines visual and textual features using adaptive global-local feature aggregation and a gated fusion mechanism.	DGM ⁴ Dataset.	Enhances the interaction between global and local features, performs binary classification, multi-label classification, image manipulation bounding box grounding, and text token grounding effectively.	Relies heavily on transformer architectures leading to increased resource usage.	Achieved superior performance (AUC: 92.84%, ACC: 86.13%) compared to state-of-the-art methods on tasks like binary and multi-label classification, image grounding (IoUmean: 76.73%), and text grounding (F1:71.29%) using the DGM ⁴ dataset.
GAN-based Visible Watermarking [36]	GANs: proactive DeepFake detection model that embeds visible watermarks in DeepFake images using regularised GANs (FSGAN _w , StarGAN _w , AttGAN _w).	FaceForensics++, RaFD, CelebA, Celeb-DF, DeeperForensics-1.0 (DF-1.0)+.	Prevents misuse of pretrained GANs by embedding visible watermarks, Works with different GAN variants and Reduces reliance on human observers.	Training watermarked GANs with regularisation terms is resource-intensive, Embedding visible watermarks may not be ideal for applications requiring unobtrusive outputs.	Maintains image quality (SSIM 0.45) while enabling effective DeepFake detection (AUC up to 0.902) and robust watermark persistence under adversarial attacks.
GAN-CNN [51]	GANs: Generative replay technique mitigates catastrophic forgetting by replaying synthetic samples during training.	Real and Fake Face Detection dataset from Kaggle.	Robustness to catastrophic forgetting, efficient continual learning, and suitability for real-world applications.	Potential misclassifications, requires further fine-tuning for diverse datasets.	Training accuracy of 98.67% with 0.0337 loss, and F1 scores of 71% Class 1 (Fake) and 69% Class 0 (Real); performance surpasses comparable models.

In order to diagnose vascular abnormalities and to improve post-surgical outcomes and stroke prevention, Doppler

ultrasonography has been used, which helps in measuring the velocity of blood flow. This method has also been widely used

to assess fetal well-being in prenatal care, thus demonstrating its versatility. In recent years, similarly pulse oximetry has developed as an important tool for determining the oxygen saturation in blood, especially in emergency and critical care settings.

Despite these methods having significantly improved healthcare by providing reliable and non-invasive diagnostic options, they have their own limitations. Pulse oximetry aids in the demonstration of how environmental variables such as illumination and pigmentation affect the accuracy. For long-term monitoring, many of the conventional approaches lack mobility and real-time functionality. Despite these limitations, they have laid a solid foundation for the current advances in physiological monitoring, therefore emphasizing the need for discoveries that address these obstacles and accommodate various diverse populations.

B. RPPG

In the recent years, rPPG has developed as an essential method for physiological monitoring. A distinct feature is that, rPPG is based on the hemodynamic response that is associated with the changes in the volume of blood beneath the skin, unlike traditional methods such as pulse oximetry or cPPG, or invasive procedures such as catheter-based assessments. Microvascular blood flow causes minute changes in the absorption and reflection of light, rPPG uses the optical properties of skin tissue to detect these changes. With the help of the color differences that is caused by pulsatile blood flow that are retrieved using sophisticated signal processing techniques and algorithms, these changes are detected through the analysis of real-time video inputs. This functionality is used especially in situations where physical contact is impossible, such as telemedicine or during infectious disease outbreaks, where a critical step is to limit the contamination risk.

rPPG offers various advantages over traditional physiological testing procedures. Due to the non-contact capability the hazards of skin irritation, pain, and infection is reduced that is associated with contact-based instruments such as rPPG probes and pulse oximeters. It is ideal to use rPPG for long-term health monitoring without disrupting everyday activities. In order to maximize the accessibility and ease, integration of rPPG sensors into everyday gadgets like webcams, telephones, and smartwatches is done.

rPPG may be used to simultaneously monitor a variety of physiological markers, which include heart rate, respiratory rate, and heart rate variability (HRV), this offers a comprehensive evaluation that outperforms the previous methods. Due to this, rPPG is an essential tool for applications that require real-time, multidimensional analysis, which include stress monitoring, fitness tracking, and cardiovascular health assessment. In order to ensure consistent performance even under poor lighting conditions or in the presences of motion anomalies, high-resolution data from video streams is used.

This method has gained significant attention in medical applications due to its ability to monitor vital signs without

direct contact [82]. Below are a few applications of this method in the medical domain:

1) CARDIOVASCULAR MONITORING

rPPG enables continuous assessment of heart rate and blood pressure, facilitating early detection of cardiovascular abnormalities [83]. For instance, a study demonstrated the use of rPPG to estimate heart rate and blood pressure from facial videos, highlighting its potential in cardiovascular health monitoring [84].

2) RESPIRATORY RATE MEASUREMENT

rPPG has been employed to measure respiratory rates, offering a non-contact alternative to traditional methods. A hospital-based trial confirmed the accuracy of rPPG in remotely measuring respiratory rate, underscoring its applicability in clinical settings [85].

3) STRESS AND EMOTION DETECTION

By analyzing variations in blood flow, rPPG can assess stress levels and emotional states, providing valuable insights into mental health. This capability is beneficial in psychiatric evaluations and stress management interventions [86].

4) SLEEP MONITORING

rPPG facilitates unobtrusive monitoring of sleep patterns, aiding in the diagnosis and management of sleep disorders. Its application in sleep studies offers a less invasive approach compared to traditional polysomnography [87].

With the shift in the healthcare paradigm to remote, non-invasive therapies, rPPG is poised to change physiological monitoring. It offers a realistic alternative to traditional approaches by allowing for cost-effective, scalable, and user-friendly diagnostics, thereby addressing limitations in portability, accessibility, and multimodal data collection.

These applications highlight the versatility of rPPG, making it a promising tool not only for healthcare but also for advanced detection systems such as deepfake forensic.

C. METHODOLOGY

The rPPG approach includes several key stages: signal capture, preprocessing, extraction, and analysis.

Figure 1 depicts the complete processing pipeline for heart rate estimation using rPPG, as described in [88] and [89]. This pipeline is employed during the post-training inference stage to process new video input and estimate heart rate. Signal acquisition is the process of recording a subject's face or other exposed skin regions with a video camera, usually under regulated lighting conditions. Regulated illumination reduces the impact of external light variations, ensuring that the data collected mainly represents physiological changes. The camera's red, green, and blue (RGB) channels are analysed to discover minor variations in skin color associated with blood volume fluctuations during the cardiac cycle [89].

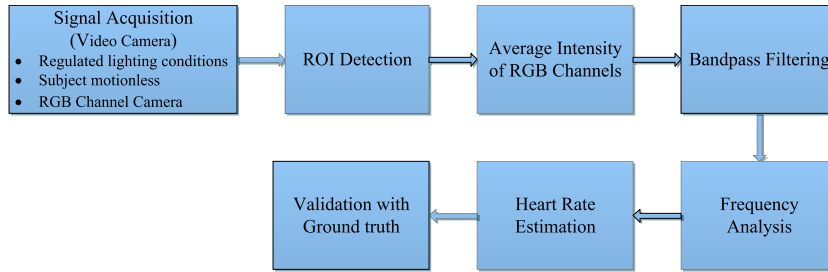


FIGURE 1. Heart rate estimation pipeline and evaluation in rPPG analysis.

During the preprocessing step, facial landmark identification algorithms, notably MediaPipe, are used to identify Regions of Interest (ROIs), such as the forehead or cheeks [88], [90], [91], [92]. Following the identification of these ROIs, the average intensity values of the RGB channels are determined for each frame [91]. This aggregation reduces random noise and highlights general color changes related to blood flow [88]. The RGB signals are modified to minimize distortions caused by mobility and ambient lighting fluctuations, resulting in signals that primarily reflect physiological variations.

Signal extraction involves dividing continuous RGB signals into shorter intervals, usually lasting 10 seconds, for efficient analysis [92]. Each portion is bandpass filtered to find the frequency components that correspond to physiological signals. The frequency range is carefully chosen to maintain components related to heart rate (0.7-3 Hz) and respiration rate (0.1-0.5 Hz), while also removing low-frequency drift and high-frequency noise.

The pre-processed signals are transferred to the frequency domain using methods like Welch's Power Spectral Density Estimation or the Fast Fourier Transform (FFT) [93]. This adjustment makes it easier to determine the prominent frequencies linked with physiological factors. According to studies, the green channel has the most correlation with variations in blood volume, and its most important peak in the frequency spectrum is used to measure heart rate [91], [93].

The classification of different rPPG techniques in Figure 2 depicts the various techniques used in remote Photoplethysmography as illustrated in [88], [90], and [91]. Signal extraction methods and procedures for providing robustness against the motion and lighting circumstances fall into two basic categories. Signal extraction methods are further classified as traditional signal processing techniques and machine learning or deep learning-based methodologies. Conventional methods, like Independent Component Analysis (ICA) [94], PCA [95], and the Chrominance-based rPPG (CHROM) method [96], are well-known for their simplicity and computational efficiency. Machine learning and deep learning techniques, such as supervised learning [45], Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs), provide adaptability and precision when extracting physiological signals [97].

For robustness, the Motion Compensation and Lighting Compensation approaches are critical in minimising environmental and subject-induced fluctuations. Adaptive filtering and color space Transformation are examples of lighting compensation techniques, while optical flow [98] and face tracking fall under Motion Compensation Techniques. These strategies improve the dependability of rPPG in dynamic and uncontrolled situations.

PCA reduces dimensionality by transforming data into uncorrelated principal components. The steps are given in Equations 5, 6, and 7 as described in [95].

$$\mathbf{C} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X} \quad (5)$$

where $\mathbf{X}$ is the mean-centered data matrix, and n is the number of observations.

$$\mathbf{CV} = \mathbf{V}\Lambda \quad (6)$$

where $\mathbf{V}$ is the matrix of eigenvectors, and Λ is the diagonal matrix of eigenvalues.

$$\mathbf{Z} = \mathbf{XV} \quad (7)$$

where $\mathbf{Z}$ is the projected data matrix.

ICA separates mixed signals into independent components. The steps are given in Equations 8 and 9 of [94].

$$\mathbf{X} = \mathbf{SA} \quad (8)$$

where $\mathbf{X}$ is the observed signals matrix, $\mathbf{S}$ is the matrix of independent components, and $\mathbf{A}$ is the mixing matrix.

$$\mathbf{S} = \mathbf{XW} \quad (9)$$

where $\mathbf{W}$ is the separating matrix.

Table 3 presents a complete overview of signal extraction methods, distinguishing between traditional signal processing approaches and machine learning-based methods [1], [21], [22], [93]. Each strategy is assessed using its strengths, limitations, and usual performance indicators. Classical approaches like ICA and PCA excel at computational efficiency and interpretability, but they might suffer from nonlinearities and noise. In contrast, machine learning, and deep learning algorithms, such as CNNs and RNNs, are more adaptable and accurate at extracting physiological

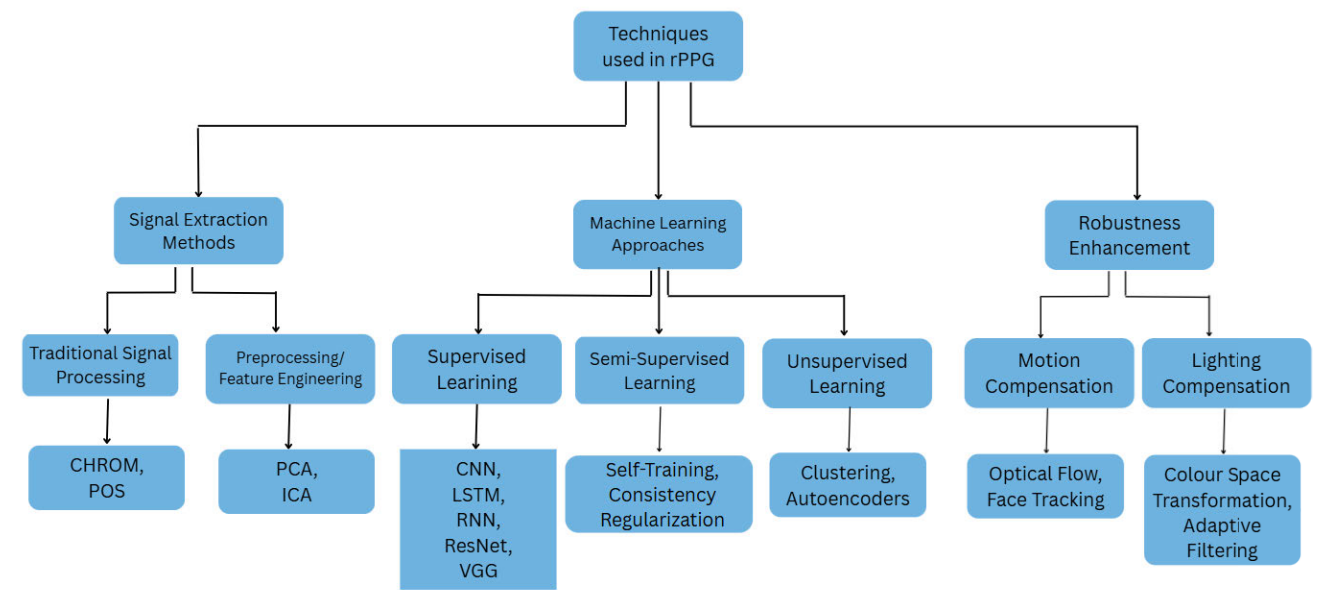


FIGURE 2. Taxonomy of rPPG techniques.

TABLE 3. Overview of techniques of rPPG.

S.No	Technique	Description	Strengths	Challenges	Performance metrics
Signal Extraction Techniques					
Traditional Signal Processing Methods					
1	Independent Component Analysis [21], [22], [94]	Separates signal into independent components, Statistical approach.	Effective isolation in static conditions, handles multiple sources.	Noise sensitivity, Motion artifacts.	Accuracy, Signal-to-Noise Ratio (SNR).
2	Principal Component Analysis [21], [22], [95]	Dimensionality reduction, Transforms data into principal components.	Simplifies data analysis, Highlights key features.	Maintaining signal integrity, Handling non-linear changes.	Efficiency, Retention, Information
3	CHROM Method [96]	Uses color information (chrominance), Extracts rPPG signals.	Effective under controlled lighting, Utilizes color data.	Accurate calibration required; Affected by varying lighting and skin tones.	Accuracy, Color Stability.
Machine Learning and Deep Learning Methods					
1	Supervised Learning [45],	Uses labeled data, Predicts signals from video.	Adaptable to various scenarios, Effective with diverse data types.	Generalizing across different environments, Variability in conditions.	Prediction Accuracy, Model Generalization.
2	Deep Learning [1]	Uses CNNs and RNNs, Learns features and patterns from video data.	High accuracy, Learns complex patterns automatically.	Computational complexity, Risk of overfitting.	Accuracy, Real-time Performance.
Robustness to Motion and Lighting Conditions					
Motion Compensation Techniques					
1	Optical Flow [98]	Tracks pixel movements to correct motion artifacts.	Effective for small to moderate movements, Real-time compensation..	Computationally intensive, Struggles with large or abrupt movements	Accuracy of motion tracking, Computational efficiency.
2	Face Tracking [21]	Uses facial landmarks to stabilise the region of interest.	Robust to facial movements, Stabilises key facial features.	Accuracy depends on algorithm precision, Can fail with occlusions.	Tracking accuracy, Robustness to occlusions.
Lighting Compensation Techniques					
1	Adaptive Filtering [93]	Dynamically adjusts filters to enhance the signal under varying lighting conditions.	Improves signal quality under fluctuating lighting, Can be tailored to specific environments.	Complex filter design, Requires fine-tuning.	Filter performance, Adaptability to lighting variations.
2	Color Space Transformation [17], [99]	Converts images to different color spaces to reduce the impact of lighting variations.	Reduces lighting variation effects, Simple to implement in certain conditions.	May lose signal information, Effectiveness depends on chosen color space.	Accuracy of color correction, Signal retention after transformation.

signals from complicated video data. Motion Compensation Techniques, such as Optical Flow and face tracking, are critical for assuring the stability of the Region of Interest (ROI) and correct signal extraction during subject motion. These solutions reduce the negative impacts of motion artifacts by dynamically adjusting ROI position and orientation.

Lighting Compensation Techniques handle the issues presented by shifting lighting conditions. Adaptive filtering and color space Transformation dynamically alter the signal processing pipeline to ensure signal integrity in varying illumination conditions. These strategies improve the signal-to-noise ratio, preserving the accuracy of derived physiological measures.

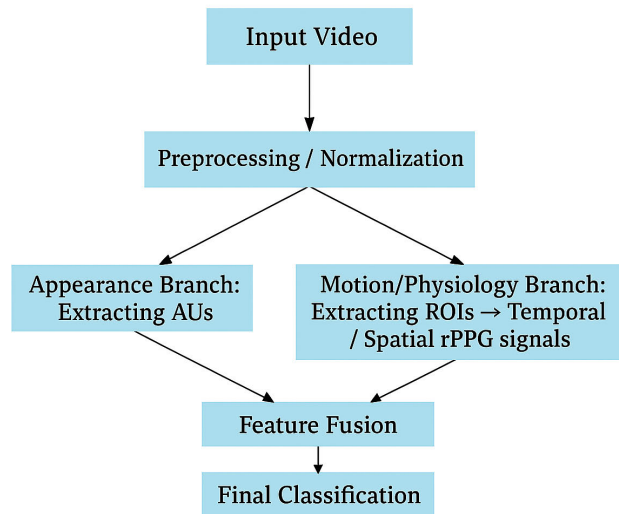


FIGURE 3. The methodology of how rPPG is used in deepfake detection showcasing its signal extraction and processing.

It also thoroughly evaluates different strategies, offering a comparative context for understanding their benefits and drawbacks.

VI. TYPES OF RPPG EXIST IN DEEPFAKES DETECTION

RPPG's ability to spot Deepfake videos is an important advance in multimedia forensics [100], which uses bodily signs to differentiate between real and fake face video data. This part examines how rPPG has evolved over time to address increasingly sophisticated Deepfake technologies [101]. RPPG looks at tiny changes in skin color, which show pulse and breathing, to develop a new way to make DeepFake detection systems more accurate and reliable.

A. METHODOLOGY FOLLOWED: SIGNAL EXTRACTION AND PROCESSING

Using rPPG in Deepfake detection requires capturing facial video data collecting physiological signals [90], [102], [103], and employing cutting-edge processing methods [104] to distinguish manipulated from true data. High-resolution video frames ensure adequate observation of minute variations in skin color caused by heartbeat cycles [105]. Certain ROIs [106] on the face—like the eyes and other parts of the face, are found for the best signal extraction. Tracking pixel intensity fluctuations throughout time inside these ROIs reveals periodic changes linked with heartbeat and breathing using dynamic signal analysis. Temporal filtering [107] and signal decomposition are advanced signal processing methods used to improve the rPPG signal and lower noise from external factors such as head motions and illumination changes. After that, the isolated signals are examined in the frequency domain for distinct trends of actual physiological activity. Last but not least, the stages of validation and calibration guarantee the dependability and correctness of the obtained signals by considering ambient variables and

personal variations [108]. This method leverages the unique capabilities of rPPG to identify DeepFake videos by utilising physiological signals to challenge fake replication. Figure 3 represents the conceptual workflow of rPPG based deepfake detection, this illustrates a parallel processing architecture as detailed in [38] and [40].

Deepfake detection's rPPG technique starts with high-resolution video frames of the subject's face. Good signal extraction depends on constant illumination and low-motion blur. ROIs and Action Units(AUs), such as the nose, cheeks, or forehead, are chosen based on where blood flow is most noticeable. These ROIs concentrate on locations with high rPPG signals, improving signal extraction accuracy [65]. Pixel intensity fluctuations inside these ROIs are monitored over time to record the periodic changes related to breathing and heartbeat.

Temporal filtering methods, such as moving average filters or adaptive filtering, improve signal extraction accuracy [109]. These methods improve the basic physiological signals and assist in reducing noise. The rPPG signal is then isolated from other noise sources using advanced signal decomposition techniques such as ICA or PCA, thereby separating the physiological signal from artifacts brought on by head movements or changes in illumination.

FFT [110], among other methods, converts the isolated rPPG signal into the frequency domain. This metamorphosis clearly shows the physiological cycles by helping to detect the unique frequency components of breathing and pulse. The peak in the frequency-transformed signal—which is related to individual heartbeats—is found using algorithms [111]. A dependable heart rate estimate depends on accurate peak detection. Furthermore, the frequency analysis points to the slower changes linked to breathing, thus offering a whole picture of the physiological signals in the video data [112].

Finally, the obtained rPPG signals are validated to guarantee their legitimacy. This entails verifying the existence of real physiological rhythms by cross-referencing with recognised physiological criteria or using control data [113]. Calibration methods offset any variances in camera settings, lighting conditions, or personal skin tone. This stage guarantees that the rPPG model offers consistent and reliable results under many conditions, strengthening the DeepFake detection system.

B. COMPARISONS OF THE MODELS RESULTS

As shown in Table 4 DeepPhys demonstrated a strong ability to identify physiological signals in face video. With a mean absolute error (MAE) of 1.2 beats per minute (bpm) for heart rate and 0.5 breaths per minute (bpm) for respiration rate, the model efficiently recovered heart rate and respiration rate information with high accuracy. The model's resilience in real-world situations was shown by constant performance under several lighting conditions and facial expressions. DeepPhys, convolutional especially in

TABLE 4. Overview of deep learning models for remote physiological measurement from video frames.

Model & Study	Description	Training Datasets	Advantages	Challenges	Results
DeepPhys: Chen et al. [114]	Utilizes a CNN with attention mechanisms to estimate heart rate and respiration rate from facial videos using rPPG signals.	MAHNOB-HCI, COHFACE	High accuracy in estimating HR and RR; robust to lighting and expression variations	Performance drops in very low-light or extreme head pose scenarios	MAE of 1.2 bpm for HR and 0.5 bpm for RR.
DeepFakesON-Phys: Hernandez-Ortega et al. [38]	DeepFakesON-Phys uses a Convolutional Attention Network (CAN) to analyze spatial-temporal video features and heart rate signals from rPPG.	Celeb-DF, DFDC	High accuracy, uses physiological cues, robust to common manipulations	Computationally intensive, affected by lighting and motion	On Celeb-DF and DFDC Preview datasets, it achieved AUCs of 99.9% and 98.2%, respectively, demonstrating superior detection performance
Simple rPPG-based Method: Giuseppe Boccignone et al. [39]	Introduced a simple rPPG-based method with the Agent in Context (AIC) and deep learning model to analyse heart rate dynamics.	FaceForensics++	Sequential floating forward selection, Feature Selection and Classification procedures	Signal Quality Variability, Computational Complexity, Generalizability, and Dataset Limitations	Achieved accuracy of up to 94.45% with an average AUC of 89.75% in Cross-method testing.
PulseEdit: Chen et al. [115]	Modifies physiological signals in facial videos to protect privacy by removing or altering heart rate data without changing the visual appearance using rPPG.	HKBUMARsV1+, Celeb-DFv1, UBFC-RPPG, PURE	Comparison over multiple rPPG extraction techniques, modification of rPPG signals, good accuracies with multiple datasets	Real-time application and diverse facial features	Achieves the highest PSNR 52.83 dB on UBFC-RPPG and 57.81 dB on PURE.
Local attention and long-distance interaction: Wu et al. [116]	Combines Mask-Guided Local Attention and a Temporal Transformer to capture spatial and temporal inconsistencies in rPPG maps for deepfake detection	FaceForensics++, Celeb-DF	Long-Distance Interaction Modelling, Signal Fusion, Local Attention Mechanism	Integration complexity and Signal quality with the extraction of rPPG signals	It outperformed with 99.55% detection for DeepFakes and 98.33% for NeuralTextures on FaceForensics++ and Celeb-DF
DeepRhythm: Qi et al. [40]	Analyses heartbeat rhythms in facial videos using rPPG and a dual-spatial-temporal attention mechanism to adapt to dynamic changes	FaceForensics++, DFDC-preview	High accuracies, Prioritising ROIs with attention layers	Computational Complexity, Generalizability, and Dataset Limitations	DeepRhythm achieves 98.0% accuracy on the ALL dataset.
FakeTransformer: Sun et al. [117]	Amplifies facial pixel variations with Eulerian Video Magnification, creating spatial-temporal maps processed by a vision transformer to identify physiological inconsistencies	FaceForensics++, DeepFake	High accuracies, Robust Temporal Analysis, and Scalability	Sensitive to video quality, high computational cost, Generalizability, Data Dependency	FakeTransformer achieves 98.98% on Deepfakes, 98.50% on Face2Face, 98.49% on FaceSwap
FakeCatcher: Ciftci et al. [37]	Analyses disruptions in facial biological signals through rPPG, applying signal transformations and CNN	Celeb-DF, FaceForensics++, FaceForensics, and a custom dataset	High accuracy, leverages physiological signals	Computational Complexity, Generalizability, Data Dependency, Sensitivity to Video Quality and Dataset Limitations	Achieved accuracies of 96%, 94.65%, 91.50%, and 91.07% on Face Forensics, Face Forensics++, CelebDF, and a new DeepFakes dataset
RhythmNet: Niu et al. [99]	Estimates heart rate from facial videos using spatial-temporal representations processed by a convolutional network and Gated Recurrent Unit (GRU)	A custom dataset: VIPL-HR	Effective in detecting temporal inconsistencies, robust to some manipulations	Complex implementation, affected by motion	Achieves heart rate estimation with an RMSE of 8.14 bpm on VIPL-HR, 8.28 bpm on MAHNOB-HCI, and 7.33 bpm on MMSE-HR
ETA-rPPGNet: Hu et al. [118]	Enhances heart rate estimation by segmenting facial video into time-domain blocks, extracting spatial features, and applying a time-domain attention mechanism to reduce noise with a CNN model	PURE, COHFACE, UBFC-rPPG, MMSE-HR	High accuracy, robust to various manipulations, utilises physiological cues	Requires high-quality video input, computationally intensive, Real-time integration	Achieves heart rate measurement with RMSE values of 4.67 bpm on COHFACE, 5.49 bpm on PURE, 5.30 bpm on MMSE-HR

noisy situations, outperformed various baseline techniques, including conventional image-based systems.

DeepPhys's success may be ascribed to its utilisation of temporal sequence analysis and deep convolutional neural networks. The model can more successfully detect minor physiological changes by using both spatial and temporal aspects from face recordings than by using stationary

image-based approaches. DeepPhys still struggles in very low-light areas and with non-frontal facial orientations, even though it works well under many circumstances. Future research might concentrate on improving its resilience in these demanding conditions by using more sophisticated image processing methods or multimodal data integration.

TABLE 5. Features used in rPPG-based deepfake detection.

Feature Type	Description	Role in Deepfake Detection
rPPG-based Temporal Patterns	rPPG detects subtle changes in skin color caused by blood flow. In deepfakes, this temporal pattern is often inconsistent due to imperfect synthesis.	Detects abnormal or missing physiological cues such as pulse rates, which are challenging to synthesise in fake videos.
Pulse Signal Amplitude	The amplitude of the rPPG signal refers to the strength of the pulse waveform, which can vary based on skin reflectance and lighting.	Deepfake videos often show irregular pulse amplitude, as the blood flow characteristics are not accurately simulated.
Pulse Rate	Refers to the heart rate, which is measurable from rPPG signals.	Deepfakes may contain incorrect or absent heart rate information due to limitations in synthesis technology.
Frequency Analysis of rPPG	Analysing the frequency domain of rPPG signals to check for biologically plausible variations, such as HRV.	Deepfake videos often fail to reproduce realistic frequency spectra in heart rates, as their generation lacks the precision to reflect normal physiological fluctuations.
Spatial Variations of rPPG	rPPG signals can vary across different facial regions (e.g., forehead, cheeks). Analysing these spatial variations can provide clues about the authenticity of a video.	In deepfakes, rPPG signals are often uniform across the face, lacking natural variations that occur in genuine videos, where blood flow may differ slightly in different facial areas.

Table 5 provides an overview of key features used in rPPG-based deepfake detection techniques. It highlights how temporal patterns, pulse amplitude, heart rate, frequency analysis, and spatial variations of rPPG signals are analysed to identify inconsistencies in synthetic videos. These features focus on detecting subtle physiological cues, which are difficult to accurately replicate in deepfake content.

Pulse Edit made major progress in generating and manipulating pulse signals from face films. Compared to groundtruth data, the model produced reasonable pulse waves with a correlation value of 0.85. PulseEdit also effectively changed pulse signals to fit desired properties, including changing the pulse rate or amplitude with minimal artifacts. The model's adaptability was shown by evaluations of its performance across many datasets, including those with varied ethnic origins and age groups.

PulseEdit's strength is its ability to change pulse signals while preserving the natural appearance of facial footage. The basic generative methodology allows perfect control over the pulse characteristics by the use of modern deep learning techniques. This feature is especially beneficial in medical education and virtual reality, where exact physiological signals are required. Still, the performance of the model may be influenced by various facial expressions and occlusions. By means of more comprehensive training sets or improved feature extraction methods, increasing the ability of the model to handle such oscillations may assist in significantly raising its relevance even more.

Furthermore, it was discovered that DeepRhythm was quite adept at identifying and examining rhythmic patterns in physical data. With a sensitivity of 0.89 and an accuracy of 0.91, DeepRhythm successfully detected patterns and consistent issues in HRV data, achieving a 92% overall success rate in differentiating between normal and aberrant rhythmic patterns. The model contained thorough visuals of rhythmic patterns, in addition to allowing one to examine body data. DeepRhythm performs well as its unique structure is designed for rhythm analysis. Mixing recurrent and convolutional neural networks captures both temporal and spectral aspects. The model may be used to monitor heart

health as it is really excellent in detecting consistent problems. Still, the incoming data's calibre and degree of detail might cause delays. If you add additional elements or utilize better data, you may notice little variations in the beat more easily. Researchers can investigate ways to mix DeepRhythm with other testing instruments going forward to obtain a better picture of cardiac conditions.

C. COMPARATIVE EVIDENCE FOR RPPG-BASED SUPERIORITY IN DEEPFAKE DETECTION

Conventional deepfake detection methods utilized visual artifacts, pixel-level features, and audio-visual synchronization to detect original and manipulated content. Early CNN and LSTM-based methods were robust, with reported accuracies in the range of 85–95% on benchmark collections like FaceForensics++ and Celeb-DF [27], [71], [74]. Multi-modal methods like EfficientNetB4 with Time-Delay Neural Networks (TDNN) are slightly better at 94.6% on DFDC [31]. These methods suffer from serious performance loss for high-quality deepfakes or adversarial compression videos since they are based on detectable pixel inconsistencies and temporal misalignments by design, which contemporary generative models and post-processing techniques remove more.

Conversely, rPPG-based detection takes advantage of involuntary physiological skin color variations resulting from blood flow as a biological watermark, hard for generative models to replicate with precision. Empirical evidence from several state-of-the-art works shows consistently better detection performance of rPPG-based models, especially on difficult cases. DeepFakesON-Phys combines convolutional attention networks with rPPG-based extraction of heart rate and achieves 99.9% AUC on Celeb-DF and 98.2% on DFDC Preview [38]. Wu et al. use mask-guided local attention with a temporal transformer to detect rPPG-driven spatio-temporal artifacts and exhibit 99.55% accuracy for DeepFakes and 98.33% for NeuralTextures [116]. FakeTransformer enhances weak rPPG signals with Eulerian video magnification before encoding with a vision transformer and attains nearly 99% accuracy for Face2Face, DeepFakes, and

FaceSwap manipulations [117]. DeepRhythm, which attacks the heartbeat rhythm coherence with dual spatial-temporal attention, attains 98% accuracy on large test sets [40].

These numerical results repeatedly surpass the performance ceiling typically reported for conventional CNN- and LSTM-based detectors [69], as well as recent multimodal systems [75], which tend to saturate at 92–95% under difficult conditions [70]. In addition, rPPG-based methods exhibit excellent generalizability to unseen types of manipulation due to the universal and involuntary nature of physiological signals. FakeCatcher [37] and ETA-rPPGNet [118] studies also show rPPG-based detection remains robust against different illuminations, minor head movement, and compression, under which artifact-based detection fails.

rPPG-based deepfake detection has been demonstrated to be more competitive or better than traditional approaches in the literature, as it provides improved accuracy and enhanced robustness to adversarial deepfakes. rPPG makes the foundation for next-generation, generalizable, and tamper-evident deepfake forensics as it acts as a physiological authentication framework.

While these results highlight the clear advantages and growing maturity of rPPG-based deepfake detection, they also undermine the need to address various challenges such as signal sensitivity to motion, lighting variations, and dataset diversity, which are discussed in the following section.

D. DATASETS SUITABLE FOR DEEFAKE DETECTION WITH RPPG

1) VIPL-HR DATASET

The VIPL-HR dataset is a comprehensive resource for rPPG research and heart rate estimation [43]. It consists of 2,378 video recordings collected from 107 participants. The dataset is designed to simulate real-world scenarios by incorporating diverse conditions, including varying illumination, head movements, and a wide range of skin tones. Additionally, the recordings were captured using multiple devices such as webcams and smartphones, allowing for cross-device validation of rPPG algorithms. To provide a reliable benchmark, the VIPL-HR dataset uses ground-truth physiological signals, such as heart rate measurements, obtained using contact-based medical devices like pulse oximeters. This dataset has proven particularly useful in evaluating the robustness of rPPG methods under challenging conditions, such as motion artifacts or fluctuating lighting. In the context of deepfake detection, the diversity and realism of the VIPL-HR dataset make it an invaluable tool for testing algorithms that rely on physiological inconsistencies to identify manipulated content.

2) MAHNOB-HCI DATASET

The MAHNOB-HCI dataset [119], originally developed for emotion recognition and human-computer interaction studies, has also become an essential resource for rPPG research. This dataset includes 527 video recordings from

27 participants and provides synchronised physiological signals, such as ECG and PPG, which serve as ground truth for heart rate and variability analysis. The recordings were conducted under controlled conditions, with participants seated while viewing emotion-evoking stimuli. Alongside the facial videos, the dataset also includes additional modalities like audio recordings and eye-gaze tracking. MAHNOB-HCI's controlled environment and high-quality annotations make it an ideal benchmark for training and evaluating rPPG algorithms. For deepfake detection, the dataset's clean and well-annotated data enables researchers to focus on identifying subtle physiological discrepancies, such as irregular heart rate variability, which are often overlooked in manipulated videos.

Together, the VIPL-HR and MAHNOB-HCI datasets offer complementary benefits. While VIPL-HR introduces real-world complexities like motion and lighting variations, MAHNOB-HCI provides a controlled environment for fine-tuning algorithms. Utilising these datasets in tandem allows researchers to build robust rPPG-based systems that can effectively detect deepfakes, even in diverse and challenging scenarios. These resources play a crucial role in advancing the reliability and scalability of physiological signal analysis in both medical and forensic applications.

VII. CHALLENGES IN DEEFAKE DETECTION USING RPPG

The current deepfake detection algorithms, particularly those that use rPPG, have few drawbacks. These challenges are divided into three categories: technical limitations, generalisation and robustness issues, and data and application challenges.

A. TECHNICAL LIMITATIONS

The detection of physiological signals utilising rPPG is extremely sensitive to ambient conditions like lighting and motion artifacts, which considerably impair the dependability of the retrieved signals. Techniques such as temporal aggregation, critical for recording physiological trends over video frames, struggle to maintain signal quality in the face of changing illumination conditions and subject mobility. The extensive temporal integration necessary for rPPG analysis creates computer constraints, particularly in dynamic situations.

Video compression, a typical storage and transmission technique, exacerbates these problems by creating chromatic distortions that obscure the tiny fluctuations in RGB channels required for rPPG signal extraction. Furthermore, optimising detection frameworks such as Neural-ODEs for resource-constrained devices is an unresolved issue, restricting the models' utility in real-world low-power systems. Motion-compensation algorithms, while promising, frequently introduce latency, limiting their usefulness in real-time systems.

One of the mitigations is adaptive illumination correction algorithms that dynamically adjust to changing illumination conditions through histogram equalisation or deep-learning-based enhancement networks. Another is multi-spectral or

near-infrared imaging to capture physiological signals in low-light environments more stably. For computational constraints, light-weight CNN architectures like MobileNetV3 or EfficientNet-lite may be employed for rPPG extraction, supplemented by hardware acceleration (e.g., FPGA or edge TPU deployment) for real-time processing. For motion artifacts, hybrid methods integrating optical flow with 3D facial mesh tracking may enable more robust motion compensation with lower latency. Another strategy could be temporal consistency regularization in training, where models are penalized for generating physiologically unfeasible variation across consecutive frames to enhance robustness against transient distortions.

B. GENERALISATION AND ROBUSTNESS ISSUES

Due to their dependence on domain-specific training data, deepfake detection systems that are based on rPPGs demonstrate restricted generalisation across a variety of datasets and manipulation techniques. Models trained on datasets such as Celeb-DF v2 and DFDC Viewing the struggle to adapt to novel settings, emphasising the domain shift issue. This problem is further complicated by adversarial vulnerabilities, in which modest perturbations introduced using techniques such as the Fast Gradient Sign Method (FGSM) or Projected Gradient Descent (PGD) drastically disturb physiological signals while preserving visual quality.

Handling low-quality, compressed, or noisy video data is another key challenge. The deterioration in spatial and temporal resolution reduces the accuracy of retrieved rPPG signals, particularly in the green channel, which is most closely connected to blood volume variations. Motion-induced aberrations cause phase errors in signal processing, which leads to inaccurate heart rate measurement. Techniques such as optical flow or facial landmark tracking offer partial solutions, but they frequently fail under complex motion scenarios or occlusions, rendering them insufficient for robust identification in uncontrolled environments.

To support generalization, domain adaptation techniques, such as adversarial learning of domain-invariant features, can be used to support models learned on one data set to generalize to other data sets effectively. Moreover, physics-based skin reflectance models of synthetic data augmentation can generate diverse rPPG signal patterns across different demographics, lighting, and motion conditions. To support adversarial robustness, defensive distillation can be used in combination with adversarial training with perturbation constraints in the rPPG domain to reduce susceptibility to small attacks while maintaining detection accuracy. Moreover, cross-modal consistency tests—where physiological signals are cross-checked with lip movements, speech rhythms, or eye-blink patterns—can serve as a defense against generative models that imitate rPPG signals but fail to coordinate them with other human activities.

C. DATA AND APPLICATION CHALLENGES

The diversity and quality of training datasets have a significant impact on the effectiveness of deepfake detection systems. However, present datasets lack demographic, environmental, and lighting diversity, resulting in biased models that perform poorly in real-world circumstances. For instance, models trained on well-lit, frontal-facing movies may fail when applied to videos with difficult angles, occlusions, or changing lighting conditions.

Real-time detection presents an additional problem since the high computing demands of processing video frames and extracting rPPG signals clash with the low-latency requirements of practical applications. Lightweight systems, while addressing delay, can cost accuracy, particularly in complicated contexts. Furthermore, the interpretability of deep learning models utilised for detection are limited. Techniques such as SHAP or Grad-CAM, while beneficial for post-hoc analysis, do not provide real-time explainability, which is crucial for high-stakes applications.

Finally, advances in generative models such as StyleGAN3 and diffusion models have resulted in the development of more sophisticated deepfakes that emit realistic rPPG signals, making detection more difficult. The progress of such technologies demands ongoing upgrades to detection models and techniques, which frequently fall behind the rate of generative model development. A particularly critical challenge arises from the fact that state-of-the-art GANs and diffusion models are increasingly capable of simulating realistic PPG and even rPPG-like signals. This directly questions whether rPPG provides a unique advantage for detection, since synthetic rPPG traces can be embedded into manipulated videos to mimic genuine physiological fluctuations. If generative models can replicate these temporal dynamics, rPPG-based detection systems risk being bypassed unless they exploit features beyond raw signal periodicity.

VIII. FUTURE DIRECTIONS

As deepfake technology continues to evolve, there are several promising directions for future research in the field of deepfake detection using rPPG:

A. ENHANCEMENT OF RPPG SIGNAL EXTRACTION

The current accuracy stands at 95%, but there is room for improvement in the future. Enhancing accuracy could involve exploring new extraction methods, such as employing advanced CNNs and leveraging state-of-the-art CNN models. Future work could also refine algorithms for more reliable rPPG signal extraction, particularly under challenging conditions like varying lighting, diverse skin tones, and motion artifacts. For instance, employing self-supervised learning techniques may enable models to learn robust signal patterns from vast amounts of unlabelled video data, so that they can handle novel situations more effectively. Adaptive band-pass filtering tuned to particular heart rate bands may also prove effective for reducing noise. Furthermore, combining

rPPG with photoplethysmography (PPG) sensors in a hybrid detection system may provide robust signals for model training.

B. INTEGRATION WITH OTHER DETECTION MODALITIES

Integrating rPPG with other modalities, such as audio-visual synchronisation, facial micro-expressions, and voice biometrics, can enhance the robustness of deepfake detection. By synchronising rPPG signals with subtle facial movements and micro-expressions, the system can detect inconsistencies that may go unnoticed when relying solely on rPPG. Similarly, analyzing voice patterns in conjunction with physiological signals can expose subtle discrepancies between the visual and auditory cues, leading to a more nuanced and precise detection framework that addresses the inherent limitations of individual methods. Multimodal transformers may be investigated to learn jointly temporal and cross-modal interactions among visual, physiological, and audio cues and enhance detection accuracy under intricate manipulations. Graph-based multimodal fusion methods may also capture relational dependencies among opposing cues to detect richer inconsistencies.

C. ADAPTATION TO ADVANCING DEEFAKE TECHNIQUES

As deepfake generation methods rapidly advance, with new models emerging every few weeks or months—such as GANs and VAEs—rPPG-based detection techniques must also evolve. To stay effective, future research can focus on developing adaptive models that can keep pace with these advancements. Leveraging adversarial training or continual learning frameworks could enable rPPG-based detection systems to learn and adapt alongside the latest deepfake creation techniques, ensuring they remain robust and effective even as deepfake technology continues to improve. Meta-learning can be used in a manner that the system can learn rapidly to adapt to new generative artifacts with minimal retraining. Bayesian neural networks can provide additional flexibility by quantifying uncertainty when new manipulations are encountered. To maintain relevance against generative models that can simulate rPPG signals, future systems must move beyond single-signal analysis. One approach is cross-modal validation, where rPPG signals are checked against other cues such as lip-synchronization, gaze dynamics, or audio prosody—areas where generative models may fail to maintain consistent physiological coordination. Another direction is to exploit subtle higher-order signal characteristics (e.g., phase relationships, inter-beat variability, and long-term coherence) that are difficult for GANs to reproduce faithfully. Continual learning frameworks can also help adapt detectors as new generative models emerge, ensuring that rPPG remains a meaningful cue even in the presence of increasingly sophisticated fakes.

D. GENERALISATION AND CROSS-DATASET VALIDATION

In order to address the generalization and robustness issues, future research can explore cross-dataset validation and

the creation of more diverse and representative datasets. Developing models that perform consistently well across various datasets and real-world scenarios will be very important for the practical deployment of rPPG-based deepfake detection techniques. Benchmarking problems that make task models struggle to fit into unseen dataset conditions may foster innovation in this field. Adding federated cross-dataset testing, where models are tested on decentralized datasets without centralizing data, can enhance robustness as well as privacy.

E. REAL-TIME DETECTION AND SCALABILITY

Another key direction is the development of real-time rPPG-based deepfake detection systems that can operate efficiently on large-scale video platforms. Future research can optimise algorithms for speed and scalability, allowing for the deployment of rPPG-based detection systems in real-time applications, such as social media monitoring and video conferencing security. Edge AI implementations could allow privacy-preserving, on-device detection, reducing bandwidth requirements while maintaining user confidentiality. Event-driven inference strategies—triggering high-complexity analysis only when anomalies are detected—could further reduce latency and computational load.

F. ETHICAL AND PRIVACY CONSIDERATIONS

As with any biometric-based technology, ethical and privacy considerations must be considered. Future work can explore the ethical implications of using rPPG signals for deepfake detection, including the potential for misuse and the protection of individuals' biometric data. Developing privacy-preserving rPPG-based detection methods will be essential for gaining public trust and ensuring responsible use of the technology. Techniques such as federated learning and homomorphic encryption could be a possible solution to train and deploy detection models without exposing the raw biometric data. Differential privacy mechanisms could also be applied to ensure that model updates do not leak sensitive information about any individual in the dataset.

Since rPPG signals constitute sensitive biometric data, information, illegal use or misuse can lead to threats e.g., identity theft, tracking, or health profiling. To overcome these issues, safe processing of rPPG data is necessary for its entire duration. Informed consent must be obtained at the collection point, storage and transmission will employ encryption, and strict access controls. Moreover, data minimization and good defined retention policies can decrease long-term exposure.

Apart from technical security, ethical deployment entails an in-depth examination of potential abuse situations, including the possibility of competitors tapping into rPPG data for unauthorized applications. Having governance frameworks, guarantee regulatory compliance, and open communication will be crucial to proper use and increasing public trust.

By pursuing these future directions, the field of deepfake detection using rPPG can continue to improve, providing more effective and reliable tools to combat the growing threat of deepfake technology.

IX. CONCLUSION

This survey explored the dynamic and challenging aspects of deepfake detection, with a particular focus on the use of rPPG techniques. As deepfake generation methods become increasingly sophisticated, traditional detection approaches often fall short of accurately identifying manipulated content. rPPG-based detection offers a promising alternative by analysing subtle physiological signals, such as blood flow-induced skin color variations, embedded within video data. However, the application of rPPG in deepfake detection is not without its challenges. Two significant hurdles include the variability in skin color and the impact of lighting conditions on the accuracy of rPPG signals. Skin color variations across different individuals can affect the strength and detectability of rPPG signals, leading to inconsistencies in detection performance. Additionally, fluctuating light conditions can distort the captured rPPG signals, further complicating the detection process.

Despite these challenges, rPPG-based methods represent a crucial step forward in the fight against deepfakes. Notably, the DeepRhythm method has achieved a significant milestone, reaching up to 92% accuracy on top datasets, demonstrating the potential of rPPG-based techniques in real-world scenarios. Future research must focus on overcoming these obstacles by developing more robust algorithms that account for diverse skin tones and adapt to varying lighting conditions. Hybrid approaches that combine rPPG with other detection techniques could also enhance the reliability and accuracy of deepfake detection.

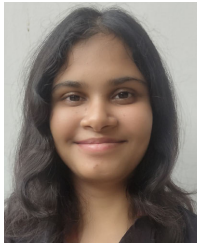
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